

Standard Process Inc.

1200 W. Royal Lee Drive
Palmyra, WI 53156

ANALYTICAL REPORT

Reported On: 17Oct2023

Eurofins Sample Code: 468-2023-07120236
Client Sample Code: HEMP OIL COMPLEX
6442-3-*-1
Sample Description: HEMP OIL COMPLEX

Industrial Hemp Cannabinoid Profile	Reference AOAC 2018.11 (modified)	Accreditation	Completed 19Jul2023	Sub 3
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Parameter	Result
Total THC (THC + (THCA x 0.877))	0.30495 mg/capsule
Cannabichromene (CBC)	0.5928 mg/capsule
Cannabichromenic Acid (CBCA)	0.0627 mg/capsule
Cannabicyclol (CBL)	<0.01425 mg/capsule
Cannabidiol (CBD)	8.265 mg/capsule
Cannabidiolic Acid (CBDA)	2.166 mg/capsule
Cannabidivarin (CBDV)	0.1083 mg/capsule
Cannabidivarinic Acid (CBDVA)	0.0399 mg/capsule
Cannabigerol (CBG)	0.4047 mg/capsule
Cannabigerolic Acid (CBGA)	0.0513 mg/capsule
Cannabinol (CBN)	0.0285 mg/capsule
Cannabinolic Acid (CBNA)	<0.00228 mg/capsule
Delta 8-THC	<0.00969 mg/capsule
Delta 9-THC	0.285 mg/capsule
Tetrahydrocannabinolic acid (THCA)	0.0228 mg/capsule
Tetrahydrocannabivarin (THCV)	<0.00456 mg/capsule
Tetrahydrocannabivarinic Acid (THCVA)	<0.00228 mg/capsule
Total Cannabinoids	12.027 mg/capsule
Total CBD (CBD + (CBDA x 0.877))	10.146 mg/capsule

The Hemp Oil Complex (Lot# 6442-3-*-1) meets the requirements of <0.3% for Delta 9 THC Levels.

Glucoraphanin	Reference Internal method	Accreditation	Completed 20Jul2023	Sub 4
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Parameter	Glucoraphanin
Glucoraphanin	8,110 µg/g

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COMPLEX 6442-3-*-1
Sample Description: HEMP OIL COMPLEX

**Pesticides <USP 561> (LC-MS/
MS and GC-MS/MS)**

Reference
USP 561

Accreditation
ISO/IEC 17025:2017
A2LA 2993.01

Completed
14Jul2023

Parameter	Result
Acephate	<0.1 mg/kg
Alachlor	<0.05 mg/kg
Aldrin and dieldrin (sum of)	<0.05 mg/kg
Azinphos-ethyl	<0.1 mg/kg
Azinphos-methyl	<1 mg/kg
Bromophos-ethyl	<0.05 mg/kg
Bromophos-methyl	<0.05 mg/kg
Bromopropylate	<3 mg/kg
Chlordane (sum)	<0.05 mg/kg
Chlorfenvinphos	<0.5 mg/kg
Chlorpyrifos	<0.2 mg/kg
Chlorpyrifos-methyl	<0.1 mg/kg
Chlorthal-dimethyl	<0.01 mg/kg
Cyfluthrin I-IV	<0.1 mg/kg
Cyhalothrin, lambda-	<1 mg/kg
Cypermethrin I-IV	<1 mg/kg
DDT (sum)	<1.0 mg/kg
Deltamethrin	<0.5 mg/kg
Diazinon	<0.5 mg/kg
Dichlofluanid	<0.1 mg/kg
Dichlorvos	<1 mg/kg
Dicofol, o,p' & p,p'-	<0.5 mg/kg
Dimethoate and omethoate (sum of)	<0.1 mg/kg
Endosulfan (sum)	<3.0 mg/kg
Endrin	<0.05 mg/kg
Ethion	<2 mg/kg
Etrimfos	<0.05 mg/kg
Fenchlorophos (sum)	<0.1 mg/kg
Fenitrothion	<0.5 mg/kg
Fenpropathrin	<0.03 mg/kg
Fensulfothion (sum)	<0.05 mg/kg
Fenthion (sum)	<0.05 mg/kg
Fenvalerate I&II	<1.5 mg/kg
Flucythrinate I&II	<0.05 mg/kg

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**Pesticides <USP 561> (LC-MS/
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Reference
USP 561

Accreditation
ISO/IEC 17025:2017
A2LA 2993.01

Completed
14Jul2023

Parameter	Result
Fluvalinate, tau- I&II	<0.05 mg/kg
Fonofos	<0.05 mg/kg
Heptachlor (sum)	<0.05 mg/kg
Hexachlorobenzene (HCB)	<0.1 mg/kg
Hexachlorocyclohexane isomers (other than gamma)	<0.3 mg/kg
Lindane (gamma-HCH)	<0.6 mg/kg
Malathion and malaoxon (sum of)	<1 mg/kg
Mecarbam	<0.05 mg/kg
Methacrifos	<0.05 mg/kg
Methamidophos	<0.05 mg/kg
Methidathion	<0.2 mg/kg
Methoxychlor	<0.05 mg/kg
Mirex	<0.01 mg/kg
Monocrotophos	<0.1 mg/kg
Parathion-ethyl and paraoxon-ethyl (sum of)	<0.5 mg/kg
Parathion-methyl and paraoxon-methyl (sum of)	<0.2 mg/kg
Pendimethalin	<0.1 mg/kg
Pentachloranisole	<0.01 mg/kg
Permethrin (sum of isomers)	<1 mg/kg
Phosalone	<0.1 mg/kg
Phosmet	<0.05 mg/kg
Piperonyl butoxide	<3 mg/kg
Pirimiphos-ethyl	<0.05 mg/kg
Pirimiphos-methyl (sum)	<4 mg/kg
Procymidone	<0.1 mg/kg
Profenofos	<0.1 mg/kg
Prothiofos	<0.05 mg/kg
Pyrethrum (total isomers)	<3 mg/kg
Quinalphos	<0.05 mg/kg
Quintozone (sum)	<1 mg/kg
S 421	<0.02 mg/kg
Tecnazene	<0.05 mg/kg
Tetradifon	<0.3 mg/kg

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6442-3-* -1
Sample Description: HEMP OIL COMPLEX

Pesticides <USP 561> (LC-MS/MS and GC-MS/MS)	Reference USP 561	Accreditation ISO/IEC 17025:2017 A2LA 2993.01	Completed 14Jul2023
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Parameter	Result
Vinclozolin	<0.4 mg/kg
The Hemp Oil Complex (Lot# 6442-3-* -1) meets the requirements for pesticides as defined by USP 561.	

Product Specification Review - Pesticides	Reference Internal method	Completed 20Jul2023
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Parameter	Result
Conformance to product specification	Pass

Heavy Metals As-Cd-Pb-Hg (Low LOQ, ICP-MS)	Reference AOAC 2013.06	Accreditation ISO/IEC 17025:2017 A2LA 2993.01	Completed 17Jul2023
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Parameter	Result
Arsenic (As)	0.33 ppm
Cadmium (Cd)	0.0057 ppm
Lead (Pb)	0.011 ppm
Mercury (Hg)	<0.0020 ppm
The Hemp Oil Complex (Lot# 6442-3-* -1) meets the requirements of heavy metals as defined by USP 2232	

Fatty Acids-Omega 6 & 3 %W/W (AOCS, Marine Oils)	Reference AOCS Ce 1b-89 mod.	Accreditation	Completed 17Jul2023	Sub
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Parameter	Result
Fatty Acid Profile, % Weight	Reported as Fatty Acids
C08:0 Octanoic (Caprylic)	<0.02 %
C10:0 Decanoic (Capric)	<0.02 %
C11:0 Undecanoic (Hendecanoic)	<0.02 %
C12:0 Dodecanoic (Lauric)	<0.02 %
C14:0 Tetradecanoic (Myristic)	0.90 %
C14:1 Tetradecenoic (Myristoleic)	<0.02 %
C15:0 Pentadecanoic	0.13 %
C15:1 Pentadecenoic	<0.02 %
C16:0 Hexadecanoic (Palmitic)	5.98 %
C16:1 Hexadecenoic (Palmitoleic)	1.47 %
C16:2 Hexadecadienoic	0.06 %
C16:3 Hexadecatrenoic	0.04 %
C16:4 Hexadecatetraenoic	0.06 %
C17:0 Heptadecanoic (Margaric)	0.15 %
C17:1 Heptadecenoic (Margaroleic)	0.10 %
C18:0 Octadecanoic (Stearic)	1.24 %

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6442-3-*-1
Sample Description: HEMP OIL COMPLEX

Fatty Acids-Omega 6 & 3 %W/W (AOCS, Marine Oils)	Reference AOCS Ce 1b-89 mod.	Accreditation	Completed 17Jul2023	Sub 1
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Parameter	Result
C18:1 Octadecenoic (Oleic)	9.80 %
C18:2 Octadecadienoic (Linoleic + isomers)	1.90 %
C18:2 Octadecadienoic Omega 6 (Linoleic)	1.73 %
C18:3 Octadecatrienoic (Linolenic + isomers)	0.40 %
C18:3 Octadecatrienoic Omega 3 (Alpha Linolenic)	0.37 %
C18:3 Octadecatrienoic Omega 6 (Gamma Linolenic)	0.03 %
C18:4 Octadecatetraenoic	0.61 %
C18:4 Octadecatetraenoic Omega 3 (Stearidonic)	0.61 %
C20:0 Eicosanoic (Arachidic)	0.17 %
C20:1 Eicosenoic (Gondoic + isomers)	5.05 %
C20:2 Eicosadienoic	0.46 %
C20:2 Eicosadienoic Omega 6	0.46 %
C20:3 Eicosatrienoic	0.22 %
C20:3 Eicosatrienoic Omega 3	0.16 %
C20:3 Eicosatrienoic Omega 6	0.06 %
C20:4 Eicosatetraenoic (Arachidonic + isomers)	1.29 %
C20:4 Eicosatetraenoic Omega 3	0.58 %
C20:4 Eicosatetraenoic Omega 6 (Arachidonic)	0.71 %
C20:5 Eicosapentaenoic	6.29 %
C20:5 Eicosapentaenoic Omega 3	6.29 %
C21:5 Heneicosapentaenoic	0.26 %
C21:5 Heneicosapentaenoic Omega 3	0.26 %
C22:0 Docosanoic (Behenic)	0.10 %
C22:1 Docosenoic (Erucic + isomers)	4.78 %
C22:2 Docosadienoic	0.02 %
C22:2 Docosadienoic Omega 6	0.02 %
C22:3 Docosatrienoic	<0.02 %
C22:3 Docosatrienoic, Omega 3	<0.02 %
C22:4 Docosatetraenoic	0.15 %
C22:4 Docosatetraenoic Omega 6	0.15 %
C22:5 Docosapentaenoic	1.31 %

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Eurofins Sample Code: 468-2023-07120236				
Client Sample Code: 91793 HEMP OIL COMPLEX				
6442-3-*-1				
Sample Description: HEMP OIL COMPLEX				
Fatty Acids-Omega 6 & 3 %W/W (AOCS, Marine Oils)	Reference AOCS Ce 1b-89 mod.	Accreditation	Completed 17Jul2023	Sub 1

Parameter	Result
C22:5 Docosapentaenoic Omega 3	0.99 %
C22:5 Docosapentaenoic Omega 6	0.32 %
C22:6 Docosahexaenoic	13.08 %
C22:6 Docosahexaenoic Omega 3	13.08 %
C24:0 Tetracosanoic (Lignoceric)	0.21 %
C24:1 Tetracosenoic (Nervonic)	0.70 %
Sum of Omega 3 Isomers	22.35 %
Sum of Omega 6 Isomers	3.48 %
Total Fatty Acids Calc.	56.97 %
Total Fat as Triglycerides	59.39 %
Total Monounsaturated Fatty Acids	21.90 %
Total Polyunsaturated Fatty Acids	26.17 %
Total Saturated Fatty Acids	8.90 %

Moulds - FDA BAM Chapter 18 mod.	Reference FDA BAM Chapter 18 mod.	Accreditation	Completed 19Jul2023	Sub 2
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Parameter	Result
Mold	< 10 cfu/g
Yeast	< 10 cfu/g

Aerobic Plate Count - AOAC 966.23	Reference AOAC 966.23	Accreditation	Completed 19Jul2023	Sub 2
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Parameter	Result
Aerobic Plate Count	10 (est) cfu/g

Escherichia coli - BAM Chapter 4	Reference FDA BAM Chapter 4	Accreditation	Completed 19Jul2023	Sub 2
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Parameter	Result
E. coli	< 3 MPN/g

The Hemp Oil Complex (Lot# 6442-3-*-1) meets the requirements for microbiological levels as defined by USP 2023 (Botanical Extracts)

Comments:
Rev.01: Units updated per client request.

Subcontracting partners:

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- 1 - Eurofins Nutrition Analysis Center, Iowa
 - 2 - Eurofins Microbiology Laboratories (Los Angeles), California
 - 3 - Eurofins Food Chemistry Testing US Madison, WI
 - 4 - Eurofins Botanical Testing US Brea, California

Respectfully Submitted,



A handwritten signature in black ink that reads 'D. Walkenhorst'.

Dana Walkenhorst
Analytical Services Manager

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